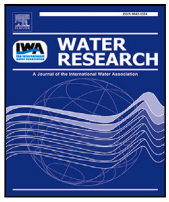
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Review



# A critical analysis of parameter choices in water quality assessment

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ABSTRACT

The determination of water quality heavily depends on the selection of parameters recorded from water samples for the water quality index (WQI). Data-driven methods, including machine learning models and statistical approaches, are frequently used to refine the parameter set for four main reasons: reducing cost and uncertainty, addressing the eclipsing problem, and enhancing the performance of models predicting the WQI. Despite their widespread use, there is a noticeable gap in comprehensive reviews that systematically examine previous studies in this area. Such reviews are essential to assess the validity of these objectives and to demonstrate the effectiveness of data-driven methods in achieving these goals.

This paper sets out with two primary aims: first, to provide a review of the existing literature on methods for selecting parameters. Second, it seeks to delineate and evaluate the four principal motivations for parameter selection identified in the literature. This manuscript categorizes existing studies into two methodological groups for refining parameters: one focuses on preserving information within the dataset, and another ensures consistent prediction using the full set of parameters. It characterizes each group and evaluates how effectively each approach meets the four predefined objectives.

The study presents that the minimal WQI approach, common to both categories, is the only approach that has successfully reduced recording costs. Nonetheless, it notes that simply reducing the number of parameters does not guarantee cost savings. Furthermore, the group of studies classified as preserving information within the dataset has demonstrated potential to decrease the eclipsing problem, whereas studies in the consistent prediction group have not been able to mitigate this issue. Additionally, since data-driven approaches still rely on the initial parameters chosen by experts, they do not eliminate the need for expert judgment. The study further points out that the WQI formula is a straightforward and expedient tool for assessing water quality. Consequently, the paper argues that employing machine learning solely to reduce the number of parameters to enhance WQI prediction is not a standalone solution. Rather, this objective should be integrated with a more comprehensive set of research goals. The critical analysis of research objectives and the characterization of previous studies lay the groundwork for future research. This groundwork will enable subsequent studies to evaluate how their proposed methods can effectively achieve these objectives.

## 1. Introduction

The water quality index (WQI) is a crucial tool in environmental monitoring, offering a comprehensive evaluation of water quality. This index transforms a variety of parameters into a single numerical value, thereby facilitating the classification of water samples into distinct safety levels (Tasneem and Abbasi, 2012; Sutadian et al., 2016; Talukdar et al., 2023; Zhao et al., 2021a). Assessing water quality through the WQI is intricately linked to the choice of parameters integrated into the formula. The utilization of specific parameter sets may categorize a water sample as safe, while alternative parameter combinations could indicate pollution within the same sample (Tasneem and Abbasi, 2012; Fernández et al., 2004). Therefore, it is important to identify the optimal parameter set to understand the quality of water.

The question of which parameter should be used to understand water quality has persisted since the development of the water quality index (Chidiac et al., 2023). Various guidelines and WQI formulas have been proposed to offer a precise approach in measuring quality of water using a range of parameters (Chidiac et al., 2023; Mogane et al., 2023). The selection of parameters for water quality assessment is influenced by the specific application or geographical context in which they are applied. Furthermore, the determination of which parameters to monitor is shaped by expert recommendations and the cost implications of collecting and analyzing water quality data (Tasneem and

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Abbasi, 2012; Chidiac et al., 2023; Mogane et al., 2023). This variation in parameter choice is known to introduce a layer of uncertainty in measuring water quality (Tasneem and Abbasi, 2012; Chidiac et al., 2023). However, with the emergence of data-driven approaches such as statistical methods or machine learning models, there is a trend towards identifying the most suitable parameter set that can address the uncertainty problem and reduce the cost of data collection (Chidiac et al., 2023; Mogane et al., 2023; Haggerty et al., 2023). Furthermore, The presence of a lengthy parameter set may lead to the eclipsing problem, where the influence of essential parameters is diminished (Uddin et al., 2021, 2022, 2023a,c; Gupta and Gupta, 2021). Identifying and focusing on the most impactful parameters becomes a critical aim in the process of parameter selection. Additionally, machine learning models have been suggested for modeling the WQI formula, leading researchers to propose the refinement of the parameter set as a strategy to enhance these models' performance (Lap et al., 2023; Bui et al., 2020; Khozani et al., 2022; Leong et al., 2021; Mohseni et al., 2024; Khoi et al., 2022).

Various data driven methodologies have been employed for parameter selection across diverse applications (Mogane et al., 2023; Uddin et al., 2023a; Fortes et al., 2023; Salem et al., 2023; Gad et al., 2021, 2020). While these methods successfully reduce the parameter set, they do not necessarily offer practical solutions for these objectives. For instance, the effectiveness of data-driven approaches in mitigating the subjectivity and uncertainty inherent in using expert opinions to select parameters remains unclear (Sutadian et al., 2016). In another example, while parameter selection is utilized to enhance machine learning models as an alternative to the WQI formula, it remains unclear how predictions of WQI are more effective than direct measurements. Consequently, a significant gap persists in reviewing previous studies and comprehensively understanding both the research objectives and the effectiveness of data-driven approaches in providing solutions. The second research gap is that data-driven methods often overlook the unique characteristics of water datasets. For instance, parameter reduction is frequently equated with cost reduction without considering the varying costs associated with different parameters. Additionally, it is necessary to demonstrate whether reduced parameters are only validated for already recorded data, or if they can also be effectively applied to future data recordings. Consequently, this study reviews previous research and responds to these dilemmas. The main objectives of this study are outlined as follows:

- This study reviews existing research on parameter selection for measuring the water quality index, categorizing it into two groups based on strategies for reducing parameter sets. It also highlights the advantages and limitations of each approach.
- This paper explores the varied motivations guiding parameter selection in water quality assessment. Although these objectives are frequently cited in the literature, this study aims to assess their validity and demonstrate how data-driven methods have effectively addressed these challenges.

Since this manuscript uses numerous abbreviations for various water parameters, a complete list of these abbreviations and their full forms can be found in Table 1. The paper encompasses several sections: It begins with a discussion on parameter selection objectives, which articulates the aims for employing parameter selection methods in the literature. This section is followed by a review of previous studies, classifying them into two groups based on their methodology. Finally, the discussion section examines various aspects of each objective, assessing how previous studies have addressed these challenges effectively.

## 2. Parameters selection objectives

In this section, four key reasons for refining parameter sets in the literature are outlined. These include reducing uncertainty, lowering costs, addressing the eclipsing problem, and enhancing the performance of machine learning models. Each objective is explored in detail to provide a comprehensive understanding of the challenges and motivations driving parameter reduction in water quality assessment.

Recognizing which parameters should be included in the WQI formula is pivotal and arguably the most crucial aspect in assessing water quality. This selection process relies on expert opinions, which can vary across different applications and geographical locations (Tasneem and Abbasi, 2012; Chidiac et al., 2023; Mogane et al., 2023). Consequently, the reliance on expert opinions is known to introduce an inherent uncertainty factor in WQI measurements (Uddin et al., 2023b; Oukil et al., 2022, 2021). This raise the question of what guidelines or principles can be employed to identify the most suitable parameters for water quality assessment through the water quality index. The majority of guidelines for water quality index formulas recommend predefined parameter sets to establish a universal standard and reduce uncertainty. For example, Soltan (Soltan, 1999) suggests a specific WQI for groundwater quality evaluation, incorporating nine parameters: nitrate, phosphate, chloride, total dissolved solids, biochemical oxygen demand, cadmium, chromium, nickel, and lead (Soltan, 1999). However, in some cases, the number and choice of parameters are left flexible, with universally applicable recommendations often missing (Sutadian et al., 2016; Kachroud et al., 2019). These guidelines

**Table 1**
Water parameters and abbreviations.

| Abbreviation | Full name |
|---|---|
| Al | Aluminum |
| Alk | Alkalinity |
| AMN | Ammoniacal Nitrogen |
| BOD | Biochemical Oxygen Demand |
| Ca | Calcium |
| Cd | Cadmium |
| Chl | Chlorophyll |
| Cl | Chloride |
| Cr | Chromium |
| COD | Chemical Oxygen Demand |
| CU | Copper |
| DIN | Dissolved Inorganic Nitrogen |
| DO | Dissolved Oxygen |
| EC | Electrical Conductivity |
| F | Fluoride |
| Fe | Iron |
| FC | Faecal Coliform |
| HCO3 | Bicarbonate |
| Hg | Mercury |
| K | Potassium |
| Mg | Magnesium |
| MRP | Molybdate Reactive Phosphorus |
| Na | Sodium |
| NH3 | Ammonia |
| NH4 | Ammonium Nitrogen |
| Ni | Nickel |
| NO2 | Nitrite Nitrogen |
| NO3 | Nitrate Nitrogen |
| Pb | Lead |
| pH | Potential of Hydrogen |
| PI | Permanganate Index |
| PO4 | Phosphate |
| Sal | Salinity |
| Se | Selenium |
| SiO2 | Silica |
| SO4 | Sulfate |
| SS | Suspended Solids |
| TC | Total Coliform |
| Temp | Water Temperature |
| TDS | Total Dissolved Solids |
| TH | Total Hardness |
| TN | Total Nitrogen |
| TON | Total Organic Nitrogen |
| TP | Total Phosphorus |
| Tran | Transparency |
| TSS | Total Suspended Solids |
| Turb | Turbidity |

recommend parameters based on expert opinions, often employing the Delphi technique (Uddin et al., 2021; Abbasi and Arya, 2000). Consequently, different sets of parameters are recommended depending on the intended purpose, whether it is for drinking, swimming, industrial use, and so forth (Uddin et al., 2021; Abbasi and Arya, 2000). Even for particular applications, there is no universal agreement on the parameters used or the WQI formula (Chidiac et al., 2023). Given the absence of a consistent consensus on parameter selection, the differences in opinions is known to introduce uncertainty into understanding water quality (Uddin et al., 2021; Abbasi and Arya, 2000; Kachroud et al., 2019).

The cost of data recording emerges as the second significant factor influencing the selection of parameters. The cost of applying expert recommendations for water quality index measurements encompasses a range of considerations, including financial constraints, labor demands, and the complexities associated with recording specific water parameters (Fortes et al., 2023; Han et al., 2022; Thakur and Devi, 2022; Paepae et al., 2021). A crucial balance is thus struck between comprehensive water quality assessment and the pragmatic aspects of cost-effective data collection, highlighting the integration of scientific rigor with economic efficiency (Fortes et al., 2023; Chidiac et al., 2023). For example, the time span between sample collection and laboratory analysis can extend for hours or even days, particularly in rural communities where the laboratory facilities are located at a distance. This time lapse can lead to chemical alterations in the samples, potentially introducing errors into the final results and impacting sampling frequency (Paepae et al., 2021). Therefore, parameters that can be swiftly assessed through portable instruments – such as temperature, potential of hydrogen, electrical conductivity, total dissolved solids, total hardness – or those that can be easily analyzed using standard laboratory equipment, like nitrate nitrogen, are preferred for inclusion in WQI measurements (Han et al., 2022; Moeinzadeh et al., 2023).

The selection of water parameters for WQI calculation is not only influenced by variations in expert opinions or the need for costly and time-consuming recording of these parameters. While one potential solution to partially address these concerns could be the development of cost-effective recording methods for a broader range of parameters, this approach may not be without its challenges and complexities. Recording a greater number of parameters may lead to the issue of parameter eclipsing in the computation of the water quality index (Ott, 1978; Smith, 1990). Parameter eclipsing is a situation where specific parameters, especially those with broader ranges or more pronounced fluctuations, exert a disproportionate impact on the final WQI calculation (Ott, 1978; Smith, 1990; Steinhart et al., 1982; Gupta and Gupta, 2021). These dominant parameters effectively overshadow the contributions of others, resulting in an imbalanced portrayal of water quality. This can be problematic because it may fail to provide an accurate reflection of the actual condition of the water sample, as the significance of other crucial parameters might be marginalized (Tasneem and Abbasi, 2012; Ott, 1978; Smith, 1990; Steinhart et al., 1982; Gupta and Gupta, 2021). Having a larger number of parameters can exacerbate this issue, as the aggregation of subindices may reduce the impact of each individual parameter, further contributing to parameter eclipsing (Tasneem and Abbasi, 2012). Addressing the parameter eclipsing issue necessitates the critical step of carefully selecting the most crucial parameters, subsequently assigning weight to them during the development of a WQI formula, and finding the proper aggregation function.

Finally, in some studies, machine learning models are utilized to predict the WQI. Consequently, various parameter selection methods are employed to feed selected parameters into the machine learning model, with the aim of enhancing its performance (Lap et al., 2023; Bui et al., 2020; Khozani et al., 2022; Leong et al., 2021; Mohseni et al., 2024; Khoi et al., 2022). Although parameters are successfully reduced in such studies, the legitimacy of replacing the WQI formula with machine learning should be assessed in conjunction with other mentioned goals. This is crucial because the WQI formula is straightforward and easy to measure, making it a practical choice. Therefore, this paper does not assess studies solely reliant on reducing parameter sets to enhance machine learning model performance.

## 3. Parameter selection approaches

The classification of water samples through the water quality index formula heavily relies on the number of parameters used (Fernández et al., 2004). Two significant factors that impact parameter selection are expert opinions and limitations in recording water's parameters. As a result, variations arise in the comprehension of water quality when distinct parameters are utilized, even within the framework of the same WQI formula (Uddin et al., 2023b; Fernández et al., 2004). The intrinsic uncertainty and variability associated with parameter selection underscore the need for the application of machine learning models or statistical methods. These tools aim to precisely identify the most relevant parameters for calculating the water quality index. The ultimate objectives are two fold: firstly, to identify parameter sets that showcase *universal* applicability across diverse WQI models, establishing independence from exclusive reliance on expert opinions (Tasneem and Abbasi, 2012); and secondly, to streamline and reduce the cost of data recording.

Water parameters undergo refinement through diverse data driven methodologies (Fortes et al., 2023; Mogane et al., 2023; Uddin et al., 2023a). These methods can be categorized into two distinct groups. In the first category, parameters are condensed using various approaches, and the refined parameters are subsequently input into machine learning classifiers to assess accuracy compared to using the full sets of parameters. Here, each sample is assigned a safety label by measuring the WQI with the full parameter sets. Subsequently, a machine learning model is trained to predict the WQI using a reduced parameter set. The fundamental concept is to introduce a framework that yields comparable classification results through the reduction of the parameter set, a concept this study terms parameter selection for consistent prediction (PSCP). In the second category, machine learning or statistical methods are utilized to retain more informative parameters while eliminating those with less variance. The WQI is then measured using the reduced parameter set (Author et al., 2011). This group of studies is referred as parameter selection for preserving information (PSPI) in this paper. A detailed list of these methods is provided in Table 3

### 3.1. Parameter selection for consistent prediction

In this category, various methods are employed to reduce parameter sets, with the goal of achieving higher WQI predictions. A particular investigation conducted a thorough comparison of eighteen different parameter selection methods applied to the same dataset, encompassing both unsupervised and supervised approaches to reduce the parameter sets (Uddin et al., 2023a). Subsequently, a deep neural network is employed as the classifier to evaluate the effectiveness of each reduced set. The findings indicate that random forest and lasso methods outperform other feature selection techniques, effectively reducing the parameter sets from eleven to eight and three, respectively (Uddin et al., 2023a).

A common approach involves the use of greedy-based parameter selection methods that iteratively choose promising parameter, aiming to construct an effective parameter set (Lap et al., 2023; Kushwaha et al., 2023; Ewaid et al., 2018). For instance, random forest in combination with recursive feature elimination is used to reduce the parameter set from ten to four (Lap et al., 2023). Similarly, the Relief method has been applied successfully, narrowing down nine parameters to six (Kushwaha et al., 2023). Additionally, another study leverages step-wise multiple linear regression (MLR), effectively reducing the parameter count from twenty-three to five (Ewaid et al., 2018).

Another group of studies investigates the statistical relationship between parameters and the measured WQI to identify redundant parameters (Ahmed et al., 2019a). Pearson correlation is used to initially

select seven parameters having high correlation with the WQI and subsequently eliminate three parameters that are highly correlated with the selected ones. This process reduces the parameter sets from nine to four. The water samples are then classified using a multilayer perceptron (Ahmed et al., 2019a).

Several studies use sensitivity analysis technique to explore various combinations of parameters, evaluating their performance by passing them to classifiers. The goal is to identify the set of parameters that achieves optimal performance to predict the WQI (Bui et al., 2020; Author et al., 2011; Asadollah et al., 2021; Ahmed et al., 2019b). Using a stepwise approach, parameters are incrementally added one at a time to a set, resulting in a total of ten parameters (Bui et al., 2020; Asadollah et al., 2021; Ahmed et al., 2019b). Through extensive exploration of various classifiers in one study, it is revealed that the parameter sets can be refined to include only five parameters (Ahmed et al., 2019b), while in another study, the optimal number is identified as three (Asadollah et al., 2021). In a similar study, optimal results were achieved with sets of four, five, and six parameters when applied to different machine learning models to estimate the WQI (Bui et al., 2020). Another set of studies employs neural networks combined with leave-one-out techniques to evaluate the relative significance of each parameter (Author et al., 2011; Othman et al., 2020; Chen et al., 2020). The analysis indicates that potential of hydrogen (pH), biochemical oxygen demand, chemical oxygen demand have less impact on neural network performance, consequently suggesting their exclusion in the WQI measurement (Author et al., 2011). Similarly, a subsequent study suggests that pH is of lesser importance in water quality evaluations (Othman et al., 2020). In a related analysis, different combinations of three parameter sets out of four are passed to various classifiers to determine which parameters are most effective in classifying water samples (Chen et al., 2020).

Selecting parameters for analysis is not solely reliant on machine learning or statistical methods; manual selection plays a significant role too. An example of this is a study where only four parameters are manually chosen to estimate an additional seven. Following this approach, the water quality index is calculated using a mix of these recorded and estimated parameters (Moeinzadeh et al., 2023).

Table 2 offers a detailed summary of prior studies in the parameter selection for consistent prediction category, specifically focusing on their efforts to reduce parameter sets and analyze the associated performance outcomes.

### 3.2. Parameter selection for preserving information

In the second category, various methodologies are employed to eliminate parameters while retaining crucial statistical information within the data. The extreme gradient boosting (XGBoost) algorithm was utilized to assess the significance of parameters (Uddin et al., 2022). The analysis revealed distinct parameter sets, with seven out of the eleven parameters were found to be significant in the summer, while five variables demonstrated significance in the winter (Uddin et al., 2022).

Another approach involves the application of statistical methods, such as principal component analysis (PCA) (Sun et al., 2016; Tripathi and Singal, 2019; Ghoderao et al., 2022; Gara et al., 2018), to diminish parameter sets. For instance, PCA is employed to downsize the parameter sets from twenty to nine by selecting parameters that convey the most information through the first five PCA components (Tripathi and Singal, 2019). Similarly, PCA is utilized in two separate studies to streamline parameter sets effectively: one study successfully reduces a set from eleven parameters to four, emphasizing the crucial insights within the first four PCA components (Ghoderao et al., 2022), while another study applies PCA to condense a set from twenty to seven, also focusing on the informative value of the first four PCA components (Gara et al., 2018). In another study, PCA and correlation analysis are used to reduce the parameter set from thirteen to five (Sun et al., 2016). In a separate research endeavor, the recording of microbiota and physiochemical parameters was undertaken with the goal of identifying physiochemical parameters highly correlated with microbiota composition (Wu et al., 2023). This effort resulted in a reduction of the parameter set from fourteen to three, clearly showcasing a correlation between the obtained WQI using these two sets (Wu et al., 2023).

Another group of studies recommends cost-effective parameters for calculating the WQI, commonly referred to as minimal water quality index ($WQI_{min}$). These studies usually compare the correlation between WQI obtained from reduced and full parameter sets to see if $WQI_{min}$ can predict the trend. Stepwise multiple linear regression is used to select a subset of five parameters from an initial pool of fifteen, focusing on those with the most significant variations within the dataset (Wu et al., 2018). In another study, five initial parameters are selected using PCA, and then stepwise multiple linear regression is applied to refine this selection to the final three parameters that effectively measure $WQI_{min}$ (Wang et al., 2023).

Few studies calculating $WQI_{min}$ select parameters based solely on the convenience of recording, without employing statistical or machine learning methods (Pesce, 2000; dos Santos Simões et al., 2008; Akkoyunlu and Akiner, 2012; Avigliano and Schenone, 2016; Kannel et al., 2007). One study demonstrates a high correlation between $WQI_{min}$, measured using five parameters, and WQI derived from fifteen parameters, indicating the feasibility of a linear transformation (Akkoyunlu and Akiner, 2012). In a different study, the recommendation is to streamline the parameter size from nine to three for the assessment of aquatic system water quality (dos Santos Simões et al., 2008). Additionally, another study compares the use of four and eleven parameters, revealing no significant difference between the values of these two indices (Avigliano and Schenone, 2016). Another separate study contrasts WQI obtained from three and twenty parameters (Pesce, 2000). It concludes that while $WQI_{min}$ is suitable for periodic or online water quality management, it falls slightly short of the accuracy achieved with the full parameter set (Pesce, 2000). Furthermore, in an independent study, a combination of five parameters is proposed as an effective indicator of the trend in the WQI, as measured using eighteen parameters. This suggests its suitability for integration into routine monitoring programs (Kannel et al., 2007). Notably, the study employs regression analysis specifically between WQI derived solely from dissolved oxygen ($WQI_{DO}$) and the WQI measured with eighteen parameters, establishing a significant correlation (Kannel et al., 2007). Consequently, both $WQI_{min}$ and $WQI_{DO}$ emerge as reliable indicators of water quality trends (Kannel et al., 2007).

## 4. Discussion

Numerous reviews have addressed the water quality index for various applications (Sutadian et al., 2016; Chidiac et al., 2023; Mogane et al., 2023; Uddin et al., 2021; Gupta and Gupta, 2021; Fortes et al., 2023; Han et al., 2022; Akhtar et al., 2021; Yan et al., 2022), alongside studies on sensors and remote monitoring techniques in WQI measurement (Jayaraman et al., 2024), and the use of machine learning in WQI modeling (Haggerty et al., 2023; Paepae et al., 2021; Zhu et al., 2022). However, a dedicated review focusing on parameter refinement techniques within WQI measurement has not yet been published. There exists a pressing need to examine the diverse objectives behind parameter selections and to explore how these goals have been addressed through machine learning or statistical methods.

It is widely known that employing various parameters introduces uncertainty into the comprehension of water quality (Tasneem and Abbasi, 2012; Zhao et al., 2021a; Uddin et al., 2021, 2023b; Kachroud et al., 2019; Abbasi and Arya, 2000; Seifi et al., 2020). This arises due to the reliance on expert opinions or the constraints in recording some parameters. Expert opinions are widely acknowledged as subjective and prone to variation based on geographical context and evolving circumstances, introducing considerable uncertainty when assessing the

**Table 2**
Overview of studies on parameter selection for consistent prediction.

| Input parameters | Algorithm | Removed parameters | Reduction ratio | Performance metric | Application |
|---|---|---|---|---|---|
| AMN, BOD, Chl, DIN, DO, MRP, pH, Sal, Temp, TON, Tran | Random Forest | AMN, Chl, Sal | 11:8 | $R^2$: 0.96 | Coastal (Uddin et al., 2023a) |
| | Lasso | AMN, BOD, Chl, DIN, DO, MRP, Sal, Temp | 11:3 | $R^2$: 0.98 | |
| BOD, COD, DO, EC, FC, NO3, pH, PO4, TS, Turb | stepwise | COD, EC, pH, PO4, TS, Turb | 10:4 | RMSE: 3.69 | River (Bui et al., 2020) |
| | | COD, pH, PO4, TS, Turb | 10:5 | RMSE: 2.76 | |
| | | pH, PO4, TS, Turb | 10:6 | RMSE: 3.65 | |
| BOD, COD, DO, NH4, PO4, TC, Temp, TSS, Turb, pH | recursive feature elimination | BOD, COD, NH4, pH, PO4, Temp | 10:4 | $R^2$: 0.90, RMSE: 11.07 | Surface water (Lap et al., 2023) |
| Ca, Cl, EC, K, MG, NA, pH, SO4, TDS, Temp, TH | manually | Ca, Cl, K, MG, NA, SO4, TH | 11:4 | F1: 0.98 | Groundwater (Moeinzadeh et al., 2023) |
| Ca, CI, EC, HCO3, K, Mg, Na, pH, SO4 | relief | HCO3, pH, SO4 | 9:6 | $R^2$: 0.99, RMSE: 10.76 | Groundwater (Kushwaha et al., 2023) |
| Al, Alk, BOD, Ca, Cl, COD, DO, EC, F, Fe, Mg, Na, NH3, NO2, NO3, pH, PO4, SiO2, SO4, Temp, TH, TDS, Turb | stepwise MLR | Al, Alk, BOD, Ca, Cl, DO, F, Fe, Mg, Na, NH3, NO2, NO3, PO4, SiO2, SO4, Temp, TDS | 23:5 | $R^2$: 0.97 | River (Ewaid et al., 2018) |
| Alk, Cl, EC, FC, NO3, pH, Temp, TDS, Turb | pearson correlation | Alk, Cl, EC, FC, NO3 | 9:4 | $R^2$: 0.74, RMSE: 2.68 | Lake (Ahmed et al., 2019a) |
| BOD, COD, DO, EC, FC, NO3, pH, PO4, TS, TU | stepwise | BOD, DO, EC, FC, NO3 | 10:5 | $R^2$: 0.94, RMSE: 2.71 | River (Ahmed et al., 2019b) |
| BOD, COD, DO, EC, NO3 , NO2 , PO4, pH, Temp, Turb | stepwise | COD, DO, EC, NO3, NO2, pH, Temp | 10:3 | $R^2$: 0.97, RMSE: 3.74 | River (Asadollah et al., 2021) |
| BOD, COD, DO, NO3, pH, SS | leave-one-out | BOD, COD, pH | 6:3 | not reported | River (Author et al., 2011) |
| BOD, COD, DO, NH3, pH, SS | leave-one-out | pH | 6:5 | r: 0.98 RMSE: 2.29 | River (Othman et al., 2020) |
| COD, DO, NH3, pH | leave-one-out | pH | 4:3 | F1: 0.98 | Surface water (Chen et al., 2020) |

**Table 3**
Overview of studies on parameter selection for preserving information.

| Input parameters | Algorithm | Removed parameters | Reduction ratio | Application |
|---|---|---|---|---|
| BOD, Chl, DIN, Do, MRP, NH4, pH, SAL, Temp, TON, Tran | XGBoost | Summer: DO, pH, SAL, TON | 11:7 | Coastal (Uddin et al., 2022) |
| | | Winter: BOD, Chl, MRP, NH4, SAL, Temp | 11:5 | |
| Alk, BOD, Ca, Cl, COD, DO, EC, FC, Mg, Na, NH3, NO3, PO4, pH, TC, Temp, TDS, TH, Turb | PCA | Alk, Ca, COD, FC, Na, NH3, NO3, PO4, Temp, TH, Turb | 20:9 | River (Tripathi and Singal, 2019) |
| Al, As, Benzene, Cd, Cl, Cr, E. Coli, EC, F, Hg, Mn, Ni, NO2, NO3, Pb, pH, Se, TC, TDS, Turb | PCA | Al, As, Benzene, Cd, Cl, E. Coli, EC, Mn, NO2, Pb, pH, TC, TDS | 20:7 | Drinking water (Gara et al., 2018) |
| DO, NH4, NO2, NO3, pH, PI, TC, Temp, TIC, TOC, TN, TSS, Turb | PCA and correlation | DO, NO2, PI, TC, TIC, TOC, TN, Turb | 13:5 | River (Sun et al., 2016) |
| BOD, Chl, COD, DO, EC, NH4, NO2, NO3, pH, PO4, Temp, TN, TP, Tran | pearson correlation | BOD, Chl, EC, NH4, NO2, NO3, pH, PO4, Temp, TN, Tran | 14:3 | Surface (Wu et al., 2023) |
| Ca, Cl, COD, DO, EC, Mg, NH4, NO2, NO3, pH, SO4, Temp, TN, TP, Turb | stepwise MLR | Ca, Cl, EC, Mg, NO2, pH, SO4, Temp, TN, TP | 15:5 | Lake (Wu et al., 2018) |
| Chl, COD, DO, NH4, NO3, pH, PO4, Temp, TN, TP, Turb | PCA and stepwise MLR | DO, NH4, NO3, pH, PO4, Temp, TP, Turb | 11:3 | Concrete water channel (Wang et al., 2023) |
| BOD, Ca, Cl, COD, DO, EC, FC, NO2, NO3, pH, PO4, SO4, TDS, TSS, Temp | manually | BOD, Ca, Cl, COD, FC, NO2, NO3, PO4, SO4, TDS | 15:5 | Lake (Akkoyunlu and Akiner, 2012) |
| BOD, DO, FC, NI, pH, Temp, TDS, TP, Turb | manually | BOD, FC, NI, pH, Temp, TDS | 9:3 | Watershed (dos Santos Simões et al., 2008) |
| DO, EC, NH4, NO2, NO3, pH, PO4, TC, TDS, Temp, Turb | manually | NH4, NO2, NO3, pH, PO4, TDS, Temp | 11:4 | River (Avigliano and Schenone, 2016) |
| NH4, BOD, Ca, CL, COD, DO, TH, Mg, NO3, NO2, Oil, pH, PO4, TDS, NA, Surfactants, Temp, Col, Turb | manually | NH4, BOD, Ca, CL, COD, TH, Mg, NO3, NO2, Oil, pH, PO4, NA, Surfactants, Temp, Col | 20:3 | River (Pesce, 2000) |
| BOD, Ca, Cl, COD, DO, EC, Mg, NH4, NO2, NO3, pH, PO4, SO4, TDS, Temp, TH, TP, TSS | manually | BOD, Ca, Cl, COD, Mg, NH4, NO2, NO3, PO4, SO4, TH, TP, TSS | 18:5 | River (Kannel et al., 2007) |

water quality index (Seifi et al., 2020; Uddin et al., 2021, 2023b; Ocampo-Duque et al., 2013). Consequently, there lacks a universal consensus on which parameters are suitable for diverse applications. This inherent complexity in defining appropriate parameters hampers the development of standardized protocols for water quality assessment. The dynamic nature of water systems and the multitude of factors influencing water quality further contribute to the challenge of establishing a one-size-fits-all approach (Pesce, 2000; Kannel et al., 2007; Chang, 2005; Penha et al., 2016; Banerjee and Ganguly, 2023; Han et al., 2019; Tirado-Conde et al., 2019; Neff et al., 2020; Munar et al., 2018; Zhang et al., 2019). Furthermore, the careful selection of effective parameters can play a pivotal role in preventing the overshadowing issues associated with the eclipsing problem (Tasneem and Abbasi, 2012). In this context, an examination will be undertaken on how these three challenges have been endeavored to be addressed by past studies, and how their solutions can effectively contribute to the future measurement of water quality indices.

The discussion is divided into two subsections. Initially, the legitimacy of the two primary goals of parameter selection is explored. Later, an analysis is conducted on how previous studies have effectively selected optimal parameter sets for their specific applications.

### 4.1. Parameter selection rationale

In this section, the prevalent notion that varying opinions contribute to uncertainty is explored. Furthermore, the utilization of machine learning models as an alternative to identify optimal parameter sets is examined. In the second subsection, the various interpretations of cost associated with the use of machine learning models are analyzed.

#### 4.1.1. Expert opinions and uncertainty

Expert opinions on determining critical water parameters are significantly influenced by the data's geographical location, local environmental conditions, associated costs, and the specific purpose of water use (Uddin et al., 2021; Zhao et al., 2021b; Hamid et al., 2020; Cheng et al., 2022). This variability is mirrored in the application of data driven methods, with the resulting discrepancies in optimal parameter identification being clearly documented in Tables 2 and 3. Such divergences and even contradictory results are evident in the literature when it comes to data analysis (Uddin et al., 2023a; Bui et al., 2020; Wang et al., 2021; Brooks et al., 2016). For instance, different sets of parameter are identified by employing eighteen machine learning and statistical methods on a specific dataset (Uddin et al., 2023a). Moreover, while spatial and temporal variations in water parameters are recognized as influential factors in water quality assessment across diverse environments (Pesce, 2000; Kannel et al., 2007; Chang, 2005; Wang et al., 2021), other research, such as the study by Penha et al. (2016), suggests that these parameters remain relatively stable across different seasons. Another crucial factor cause differences in water's parameter trend is the intricate interplay between different water types and their influence on quality (Penha et al., 2016; Banerjee and Ganguly, 2023; Han et al., 2019; Tirado-Conde et al., 2019; Neff et al., 2020). For instance, groundwater and surface water can mutually impact each other's quality (Penha et al., 2016; Banerjee and Ganguly, 2023). Temporal variations in water parameters within rivers fed by springs may vary, especially during the dry season. Conversely, the river can actively contribute to the recharge of aquifers, though with the potential contribution to water pollution in the groundwater. Penha et al. (2016), Banerjee and Ganguly (2023). Additionally, anthropogenic activities significantly contribute to nutrient loading, which detrimentally affects both water bodies and aquatic ecosystems (Dorado-Guerra et al., 2023; Mayor et al., 2023). Moreover, climate change further amplifies these complexities, with the potential to significantly influence the hydrological processes of watersheds and consequently impact the hydrodynamics and quality of their water bodies (Munar et al., 2018; Zhang et al., 2019). While the diversity in expert opinions reflects a subjective interpretation of each case study, it seems justifying results from data driven methods remains challenging without a study-specific subjective analysis.

Analyzing the dynamic changes in water parameters requires an individualized examination for each application, employing diverse parameter types to understand the optimal parameter set for particular application. Attempting to establish a universally applicable set of parameters seems impractical, given the varied and complex nature of different water systems and their unique environmental influences. Even if experts were to reach a consensus on a universal parameter selection, the result would likely be an extensive array of parameters attempting to address the varying needs of different conditions. This approach would not only be cost-prohibitive but also impractical to implement. Therefore, the diversity in expert opinions across different conditions not only avoids introducing any uncertainty but should also be regarded as advantageous. This variety underscores the necessity for thoughtfully selected parameters, tailored to specific applications and the continually changing environmental conditions.

Physical-driven modeling stands as a valuable approach for comprehending the intricacies of hydrological systems, and predict changes and importance of water parameter over period of time. This approach focuses on the physical aspects of water flow's internal movement to monitor how pollutants vary spatially and temporally (Tan et al., 2023). It provides a systematic means to investigate an array of factors, encompassing climate change, weather patterns, soil characteristics, and the dynamic movement of water within and between watersheds and water bodies. This approach not only enhances our current understanding of hydrological processes but also contributes significantly to anticipating and understanding the potential changes in water parameters in the future (Tan et al., 2023; Ejigu, 2021). While challenges such as the difficulty in quantifying certain processes, constraints imposed by climate and environment exist, along with the requirement for extensive field data, physical-driven modeling remains a potent tool for estimating future changes in water parameters (Tan et al., 2023; Ejigu, 2021; Read et al., 2019).

When transitioning from expert opinions to machine learning models, machine learning offers the advantage of assessing various factors and utilizing the historical data of water quality changes to determine which parameters can effectively represent water quality. Additionally, it is not constrained by the limitations inherent in physically-driven modeling, leading to a more straightforward implementation process. Traditionally, machine learning models or statistical methods have been used mainly to identify the best parameters for specific datasets, a process that still relies heavily on expert judgment (Sutadian et al., 2016).

The main aim of machine learning in this context has been to refine and minimize the parameter set, rather than to independently determine the most effective parameters that fully capture water quality for a particular use. Factors like cost, recording time, and environmental variations have not been the primary focus. Essentially, machine learning or statistical methods have served more as a tool to streamline parameter selection based on existing data, rather than replacing expert opinions. Its key contribution has been in potentially reducing the costs associated with data collection by limiting the number of parameters that need to be recorded.

#### 4.1.2. Cost dynamics of parameter selection

The term cost is not clearly defined in previous studies, and it is generally assumed that reducing the number of parameters correlates with a decrease in costs. However, the actual cost is contingent upon the methodology of water parameter sampling and measurement. Traditionally, it involves a trained individual traveling to the water source, collecting water samples, and returning them to a laboratory for subsequent analysis (Paepae et al., 2021). Various factors contribute to the overall cost, including the availability of techniques, the time

elapsed from sample collection to laboratory analysis, expenses associated with parameter analysis, and the precision of data analysis (Thakur and Devi, 2022; Paepae et al., 2021). The development of new sensors has made it more cost-effective to record numerous parameters and increase the sampling frequency (Thakur and Devi, 2022). The utilization of wireless sensor networks further aids in reducing the costs associated with real-time monitoring (Demetillo et al., 2019). However, it is important to note that these approaches may introduce additional expenses, such as higher power requirements, vulnerability to cyber-attacks, elevated installation and maintenance costs, as well as potential storage issues (Paepae et al., 2021). Hence, the cost of measurement is highly dependent on the methodology used for data collection and analysis.

Previous studies neglect to provide detailed descriptions of their data collection methods, rendering it difficult to assess the impact of parameter reduction on recording costs. Furthermore, there is a lack of research focused on evaluating the cost implications of recording and identifying which parameters are costly based on their measurement techniques. Consequently, despite the frequent promotion of cost reduction as a significant advantage of applying machine learning or statistical methods, most study has no specifically aimed to eliminate costly parameters. The prevailing assumption is that removing any parameter automatically leads to cost savings, without a targeted focus on the cost implications of each parameter. This assumption persists even though several of the excluded parameters (as mentioned in Tables 2 and 3) are actually straightforward and cost-effective to measure. Notably, some studies employing minimal water quality index $WQI_{min}$ strategies have utilized parameters that are known for their ease and low cost of recording, indicating a practical approach to achieving cost efficiency while maintaining reliable water quality assessment (Moeinzadeh et al., 2023; Pesce, 2000; dos Santos Simões et al., 2008; Akkoyunlu and Akiner, 2012; Avigliano and Schenone, 2016; Kannel et al., 2007).

Another cost-related aspect is the errors associated with using reduced parameter sets. It must be acknowledged that an error arises from the difference between employing a complete set of parameters versus a reduced one. In other words, the inquiry seeks to understand the consequences of diminishing parameter sets on the water quality index. This error is typically reported in studies categorized under parameter selection for consistent prediction. Meanwhile, in studies in parameter selection for preserving information category, alterations to the WQI formula or weighting factors are common, which complicates the understanding of potential misclassifications in water samples when using reduced parameter sets (Pesce, 2000; dos Santos Simões et al., 2008; Akkoyunlu and Akiner, 2012; Avigliano and Schenone, 2016; Kannel et al., 2007).

### 4.2. Challenges and perspectives

In this section, the investigation is conducted to determine whether the selected parameter set can be reliably and effectively substituted for the complete set. Essentially, the goal is to assess whether data driven methods offer compelling evidence that the exclusion of certain parameters does not significantly impact water quality measurements. Consequently, the aim to establish its compatibility with the original recommendations from the experts.

A set of studies, categorized as parameter selection for consistent prediction, employs diverse techniques for the initial reduction of parameter sets. Subsequently, machine learning models are utilized to predict WQI labels or values using the reduced set. Therefore, the selected parameters are exclusively recommended for input into a machine learning model and are not intended for use in the WQI formula. As a result, systematic comparison is conducted between the performance of their models with the use of full parameter sets and those using the reduced set. While this approach still relies on parameters suggested by experts and does not offer a solution for the eclipsing problem, it remains a valid method to streamline parameter sets, acknowledging the ongoing trade-off between the expected errors in machine learning models and the cost of recording removed parameters.

The second set of studies, referred to as parameter selection for preserving information, advocates for the utilization of the reduced parameter set in the WQI formula. Consequently, the removed parameters are asserted to have no impact on the measured WQI. It is important to underscore that some studies employ different weighting factors or aggregation functions for the reduced parameter sets (Uddin et al., 2022; Tripathi and Singal, 2019; Wu et al., 2018). These alterations have an impact on the final water quality index value when compared with the full parameter sets. Consequently, the final reported results do not solely reflect the impact of reduced parameter sets. In this context, the discussion is not extended to explore the effects of other factors in the WQI formula.

As depicted in Table 3, various statistical, machine learning models, or even manual approaches are employed for parameter refinement. It is important to highlight that statistical methods, often sensitive to parameter variability, may introduce bias in parameter selection. For instance, principal component analysis and pearson correlation rely on parameter variance to identify meaningful patterns in a dataset (Maćkiewicz and Ratajczak, 1993; Faizi and Alvi, 2023). The PCA employs the variance of different parameters to identify influential factors, emphasizing those contributing significantly to dataset variability (Maćkiewicz and Ratajczak, 1993). This method facilitates the selection of key features by capturing dimensions with the highest variance. Similarly, pearson correlation gauges the strength and direction of linear relationships between parameter pairs based on variances (Faizi and Alvi, 2023). A high parameter variance implies increased potential for correlation analysis to unveil meaningful associations. Although such methods can reduce parameter sets, their ability to ensure the exclusion of less significant parameters is not guaranteed. Various scenarios can illustrate this condition. For instance, as was demonstrated earlier, lead, cadmium, chromium, and copper are shown to exhibit sparse values in the dataset used, causing them to be indicated as having high mutual information with other parameters (Moeinzadeh et al., 2023). In another scenario, even parameters with lower variability may display values exceeding permissible limits, potentially signaling water pollution. This occurs despite these parameters being candidates for removal from the parameter set. Another example is the exclusion of dissolved oxygen, a crucial factor for understanding water quality and essential for the survival of all aquatic organisms (Markfort and Hondzo, 2009; Cox, 2003; Kannel et al., 2007), is often excluded from parameter lists through these techniques (Uddin et al., 2022; Sun et al., 2016; Tripathi and Singal, 2019; Pesce, 2000). Such exclusions could potentially lead to a misunderstanding of water quality, as evidenced by the proactive step of manually reintegrating dissolved oxygen into the selected parameters after its initial exclusion by the PCA method (Tripathi and Singal, 2019). Regardless of whether methods within the PSPI category eliminate critical parameters, reducing the number of parameters involved in measuring the water quality index has the potential to mitigate the eclipsing effect (dos Santos Simões et al., 2008).

In each application utilizing the data driven methods, comprehension of the dataset is enhanced. The pivotal question becomes whether this newfound understanding can be translated into actionable knowledge for specific applications. Can reliance be confidently placed on these methods to exclude the monitoring of particular parameters? It becomes evident that relying solely on these methods for such recommendations is not feasible. This is due to our awareness that water parameters are temporal, exhibit variations across different locations, and are highly influenced by external environmental factors (Pesce, 2000; Kannel et al., 2007; Chang, 2005). Most importantly, experts recommend that every set of parameters should be carefully monitored, as they are essential for the intended application. Therefore,

it is essential to provide robust validation to demonstrate that these parameters are not important for that particular application, especially in the future data recording. Based on the points raised to revise the research objectives of parameter selection, this study proposes two future research directions:

- Machine learning holds the potential to identify more effective parameters for comprehending water quality, especially when it incorporates various influential factors on water's characteristics. This data could encompass elements like weather, geographical variations, pollution sources, seasonal changes, and human activities. In such scenarios, the parameter sets determined by machine learning might not be universally applicable but instead could be locally optimized, tailored to specific conditions to provide a clearer and more accurate representation of water quality. This localized approach to parameter optimization, a notable yet untapped potential in water quality research, has not been extensively explored in existing studies. The adaptability of machine learning allows for dynamic selection of parameters, catering to the specific needs of various environments over time. Consequently, machine learning stands as a potent alternative to traditional expert analyses, with the potential to reduce costs and address the eclipsing problem more effectively. Efficient parameter selection facilitates the creation of long-term datasets that are essential for understanding and responding to long-term environmental trends.
- Cost analysis plays a pivotal role in parameter refinement. It is vital to clearly outline the costs associated with recording each parameter along with the associated benefits and risks of their removal. Consequently, machine learning models should be strategically designed to minimize practical costs while ensuring that essential parameters critical for future recordings are preserved.

## 5. Conclusion

This manuscript reviews prior research focused on refining the selection of parameters used to calculate the water quality index. It enumerates four key objectives common to these studies: reducing uncertainty and costs, minimizing the eclipsing effect, and improving the accuracy of machine learning models in predicting WQI. The study articulates these aims and examines how previous research has addressed them. Additionally, it categorizes the literature into two distinct groups based on the methodological approach to parameter selection: one aimed at ensuring consistent prediction with using full parameter set (PSCP) and the other at preserving information within the data (PSPI).

This study posits that merely reducing the parameter set to enhance machine learning model performance should not be an isolated objective; instead, it must be evaluated in conjunction with other goals. Additionally, it reveals that previous research has not effectively mitigated the uncertainty traditionally associated with the choice of parameters. The study advocates for a diverse array of expert opinions, emphasizing that they contribute positively to the adaptability to various environmental conditions. It argues against the feasibility and cost-effectiveness of a universal parameter set for assessing water quality.

The study suggests that machine learning holds the potential to effectively replace expert judgments, significantly lessening uncertainty in water quality parameter selection by accounting for diverse environmental conditions. It also explores various aspects of cost analysis, indicating that parameter reduction does not necessarily result in cost reduction. The study concludes that only the minimal WQI approach has notably decreased recording costs, acknowledging the potential accuracy trade-off associated with not employing the complete parameter set. Finally, it delineates that while studies in the PSPI category have the capacity to mitigate the eclipsing effect, those in the PSCP category cannot effectively address this issue.

Overall, this paper proposes considering various aspects of reducing parameter sets, offering a framework for future studies to evaluate their effectiveness in parameter refinement. It is essential that the outcomes of data-driven methods be validated and contextualized with existing knowledge from data recording sites. Moreover, this paper illustrates how data-driven approaches can streamline the removal of less critical parameters from previously recorded data and potentially from future data collections.

## CRediT authorship contribution statement

**Hossein Moeinzadeh:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Ken-Tye Yong:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Anusha Withana:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

## Declaration of competing interest

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## Data availability

No data was used for the research described in the article.

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